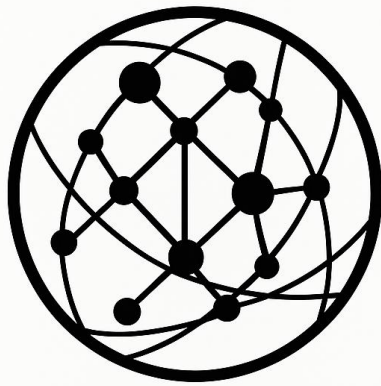


Policy Brief

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Beyond Hormuz:

Iran’s Ten-Corridor Logistics Adaptation under Blockade Pressure
Caspian Shipping, Central Asian Rail, and the Multi-Domain Logic of Threshold-Delaying Supply

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Key Judgments

- **Iran’s ten-corridor plan is announced but undisclosed.** Tehran has confirmed alternative corridors, but has not released a full route list.
- **The framework diversifies logistics; it does not replace Hormuz.** Alternative routes can sustain minimum flows, not Persian Gulf-scale throughput.
- **The Caspian Sea is the clearest northern buffer.** Its observed shipping activity supports a threshold-delaying role rather than surge capacity.
- **Central Asian rail shows readiness, not confirmed Iran-bound supply.** The data indicate usable overland capacity, but not final cargo destination.
- **Air routes add a high-value replenishment layer.** They matter for light, time-sensitive goods, not bulk trade.
- **Hormuz is degraded, not sealed.** Residual movement shifts the blockade from total denial to friction, delay, risk, and selective permeability.

Executive Summary

Iran’s reported plan to develop ten distinct alternative trade corridors marks a shift in its wartime logistics strategy. The evidence does not show that Tehran can replace the Persian Gulf and Strait of Hormuz system. Rather, it suggests that Iran is trying to turn blockade pressure from a flow-denial problem into a friction-management problem.

Because Iran has not released a full corridor-by-corridor list, this brief offers a plausible analytical reconstruction based on public reporting, geography, existing trade patterns, and observed logistics behavior. The core finding is that Iran’s alternative corridor system is not a surge-capacity network, but a threshold-delaying sustainment system designed to preserve minimum viable flows of food, medicine, industrial inputs, selected exports, dual-use-relevant goods, and time-sensitive components.

The system extends beyond land and sea corridors. Air connectivity adds a high-value replenishment layer, while residual movement through a degraded but not fully sealed Hormuz/Persian Gulf window preserves limited maritime flow. Iran’s ten-corridor plan therefore does not mean the blockade has failed; it means the blockade is becoming porous, frictional, and long-duration.

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Why This Matters

Hormuz is not only a chokepoint; it is the organizing center of Iran’s external logistics system. Once disrupted, the key question is not simply how much shipping capacity Iran loses, but how much minimum flow it can preserve through secondary, tertiary, and residual channels.

Iran’s ten-corridor plan matters because a blockaded state does not need full logistical recovery to remain resilient. It only needs enough continuity to slow depletion, sustain basic consumption, preserve selected industrial inputs, and keep high-value replenishment routes open, thereby slowing the conversion of blockade pressure into systemic collapse.

1. The Public Claim: Ten Corridors, No Full Map

Iranian officials have publicly discussed the need to define and map alternative trade routes under conditions of war and blockade. Public reporting points to a broader ten-route framework, while separate reporting indicates that six alternative routes have already been defined. The apparent difference likely reflects a distinction between routes already defined for near-term use and a wider set of identified or planned alternatives.

The public record, however, has a major limitation: Iran has not released the full list of ten distinct alternative trade corridors. Existing reporting provides directional clues: North–South and East–West corridors, Central Asia, Eurasian routes, northern ports, railway, road, maritime options, Pakistan, and reopened older routes, but not a complete route map.

This distinction matters. This brief does not claim to reveal the official ten corridors. It reconstructs a plausible ten-corridor framework based on available evidence and evaluates how such a system would function under blockade pressure.

2. Analytical Reconstruction: The Likely Ten-Corridor Framework

The following table offers a plausible reconstruction of Iran’s ten-corridor logistics adaptation. It is based on public clues, geography, known trade linkages, observed shipping behavior, and the logic of wartime redundancy.

Because Iran has not released an official route list, the table below should be read as an analytical reconstruction rather than a confirmed corridor map.

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Table 1. Plausible Ten-Corridor Framework under Blockade Pressure

No.	Possible Corridor	Type	Primary Function	Capacity	Strategic Effect	Confidence
1	Russia–Caspian Sea–Northern Iran	Maritime	Bulk goods, grain, industrial inputs, dual-use-relevant inputs	Medium	Strong threshold-delay effect	High
2	Kazakhstan–Caspian Sea–Iran	Maritime / multimodal	Grain, raw materials, bulk cargo, transshipment	Medium	Adds northern redundancy	Medium
3	Turkmenistan–Iran Northeast	Rail / road	Central Asian access, border trade, basic goods	Low–Medium	Regional continuity channel	Medium
4	Kazakhstan–Turkmenistan–Iran railway	Rail	Containerized and bulk cargo	Medium	Reduces southern maritime dependence	Medium
5	China–Central Asia–Iran rail-linked route	Rail / multimodal	Components, machinery parts, containerized cargo	Low–Medium	Enables selective supply continuity	Medium
6	Turkey–Iran western corridor	Road / rail	Imports, exports, rerouting	Medium	Western redundancy and market access	Medium
7	Iraq–Iran western border corridor	Road	Regional trade, informal flows, transshipment	Low–Medium	Local resilience and informal-network risk	Medium
8	Pakistan–Iran southeastern corridor	Road / rail potential	Food, agricultural goods, basic commodities	Low–Medium	Emergency commodity support	Medium
9	Chabahar–Oman Gulf / Indian Ocean route	Maritime / land-sea	Southeast maritime alternative	Medium	Partial bypass of Hormuz-centered pressure	Medium
10	Afghanistan–Iran eastern corridor	Road / rail potential	Border trade, low-cost flows, Central Asia linkage	Low	Auxiliary continuity route	Low–Medium

Source: Author’s reconstruction based on public reporting on Iran’s alternative-corridor planning, regional trade geography, the author’s Caspian/Persian Gulf monitoring dataset, and China Railway Express data through March 2026.

Note: This is not an official Iranian list. The table reconstructs plausible corridors and assesses their likely roles, capacity, and strategic effects under blockade pressure.

This framework should not be interpreted as ten equal routes. It is a layered system. The Caspian and Central Asian routes form the northern backbone. Turkey, Iraq, Pakistan, and Afghanistan provide border redundancy. Chabahar provides a southeastern maritime alternative. China–Central Asia rail connectivity provides a potential enabling layer for high-value and containerized cargo.

The system is best understood as a distributed sustainment architecture, not a substitute for Persian Gulf maritime scale.

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3. Functional Assessment of the Ten Corridors

The ten reconstructed corridors differ in capacity, reliability, visibility, and strategic effect. The following assessment summarizes each corridor’s likely role and constraints under blockade pressure.

3.1 Russia–Caspian Sea–Northern Iran

This is the most important and observable alternative corridor. It links Russian Caspian ports with northern Iranian ports such as Amirabad and Anzali, supporting grain, industrial materials, machinery, selected dual-use-relevant inputs, and other high-priority goods. Its main constraint is node concentration: port capacity, berth availability, vessel size, shallow-water conditions, cargo handling, and security all limit scalability.

Assessment: high strategic value; medium capacity; high monitoring priority.

3.2 Kazakhstan–Caspian Sea–Iran

Kazakhstan provides a secondary Caspian-facing option for grain, raw materials, bulk commodities, and transshipment cargo. Its value lies in northern diversification, but it remains dependent on Caspian port infrastructure and vessel availability.

Assessment: medium strategic value; medium capacity; useful secondary northern source.

3.3 Turkmenistan–Iran Northeast

Turkmenistan offers proximate land access into northeastern Iran through road and rail links. It can support food, fuel, construction materials, and regional trade, but border processing, customs friction, rail scheduling, and political coordination limit scale.

Assessment: medium continuity value; limited surge potential.

3.4 Kazakhstan–Turkmenistan–Iran Railway

This corridor provides a structured Central Asian rail pathway for containerized cargo, bulk goods, and industrial inputs. Rail offers predictability, but transshipment delays, infrastructure bottlenecks, scheduling limits, and single-node disruptions constrain flexibility.

Assessment: important medium-volume corridor; strong redundancy value.

3.5 China–Central Asia–Iran Rail-Linked Route

This route is analytically important because available China–Central Asia rail and TEU data through March 2026 show usable regional activity. The data do not prove Iran-bound deliveries, but they indicate an enabling layer for high-value goods, components, machinery parts, electronics, and lightweight industrial inputs.

Assessment: low-to-medium capacity; high value for selective replenishment; evidence should be treated cautiously.

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3.6 Turkey–Iran Western Corridor

Turkey provides a western commercial interface connected to Europe, the Mediterranean, and overland Eurasian routes. It can support imports, exports, and rerouting, but is more politically visible and exposed to customs scrutiny, sanctions pressure, and Western-aligned monitoring environments.

Assessment: medium capacity; high commercial value; politically visible.

3.7 Iraq–Iran Western Border Corridor

The Iraq–Iran border corridor can support regional trade, transshipment, informal flows, and small-scale continuity. It is unlikely to provide high-volume replacement capacity, and its risks include border-security constraints, informal-network exposure, customs weakness, and high monitoring sensitivity.

Assessment: low-to-medium capacity; useful for local resilience; high security risk.

3.8 Pakistan–Iran Southeastern Corridor

Pakistan is relevant for basic commodity supply, especially food, rice, meat, and agricultural goods. Its value lies in food-security stabilization rather than maritime replacement, while infrastructure, border processing, and security constraints limit scale.

Assessment: medium relevance for basic commodities; limited industrial capacity.

3.9 Chabahar–Oman Gulf / Indian Ocean Route

Chabahar offers a southeastern maritime option outside the narrowest Hormuz-centered frame and can connect Iran to Indian Ocean trade. Its value is conditional, however, because it remains exposed if enforcement applies broadly to Iranian ports.

Assessment: medium potential; high strategic symbolism; exposed under comprehensive blockade.

3.10 Afghanistan–Iran Eastern Corridor

The Afghanistan corridor is the weakest in capacity but can still support border trade, low-cost flows, and links to Central Asia. Its role is auxiliary, limited by security, governance, road quality, and political uncertainty.

Assessment: low capacity; auxiliary continuity route.

4. Beyond the Ten Corridors: Air Replenishment and Residual Hormuz Flow

The ten-corridor framework does not capture the full scope of Iran’s logistics adaptation. Two additional layers matter: air routes and residual maritime movement through Hormuz.

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Air connectivity should not be treated as one of the officially reported ten corridors unless Iran explicitly identifies it as such. Analytically, however, aviation forms a high-value, low-volume replenishment layer. Restored or partially restored Iran–China air connectivity does not prove movement of specific cargo, but it does show that air access was being restored while maritime routes remained contested. In a constrained logistics system, aviation can matter disproportionately for electronics, medical goods, precision components, machine parts, communications equipment, and lightweight industrial inputs.

Hormuz should also not be analyzed as a binary open-or-closed chokepoint. Even under blockade pressure, residual maritime movement may continue through contested windows. This does not mean Hormuz is normal. It means Hormuz is degraded but not fully sealed. A blockade can remain effective even if it leaks, provided it reduces throughput, raises cost, delays delivery, increases risk, and disrupts cargo prioritization.

5. A Three-Layer Logistics Adaptation Model

Iran’s current adaptation can be represented as a three-layer system.

Table 2. Iran’s Multi-Domain Logistics Adaptation System

Layer	Component	Function	Capacity	Strategic Role
Layer 1	Ten land–sea alternative corridors	Basic goods, bulk cargo, border trade, selected exports	Low–Medium	Maintains minimum viable trade
Layer 2	Air replenishment routes	High-value, low-volume, time-sensitive cargo	Low	Preserves critical inputs
Layer 3	Residual Hormuz/Persian Gulf flow	Limited continuation of core maritime movement under risk	Degraded but nonzero	Prevents complete maritime cutoff

Source: Author’s analysis based on the reconstructed corridor framework in Table 1 and observed logistics indicators.

Note: The three-layer model is an analytical framework. It groups Iran’s adaptation into land–sea corridors, air replenishment, and residual Hormuz/Persian Gulf flow; it should not be read as an official Iranian classification.

This model clarifies why the ten-corridor plan matters. The corridors are not expected to replace Hormuz. They are expected to combine with air and residual maritime flows to delay systemic breakdown.

The result is a multi-domain, threshold-delaying supply system.

6. Evidence Check: Two Observable Corridors

Two corridors provide the strongest observable evidence for the broader framework: Caspian shipping and Central Asian rail connectivity.

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6.1 Caspian Sea corridor.

The Caspian corridor provides the strongest observable evidence of Iran’s northern logistics adaptation. Port-output and vessel-monitoring data show that the Caspian route is not merely theoretical; it has functioned as a measurable buffer under pressure. However, its volatility and limited vessel depth suggest a threshold-delaying role rather than a replacement for Persian Gulf-scale maritime logistics.

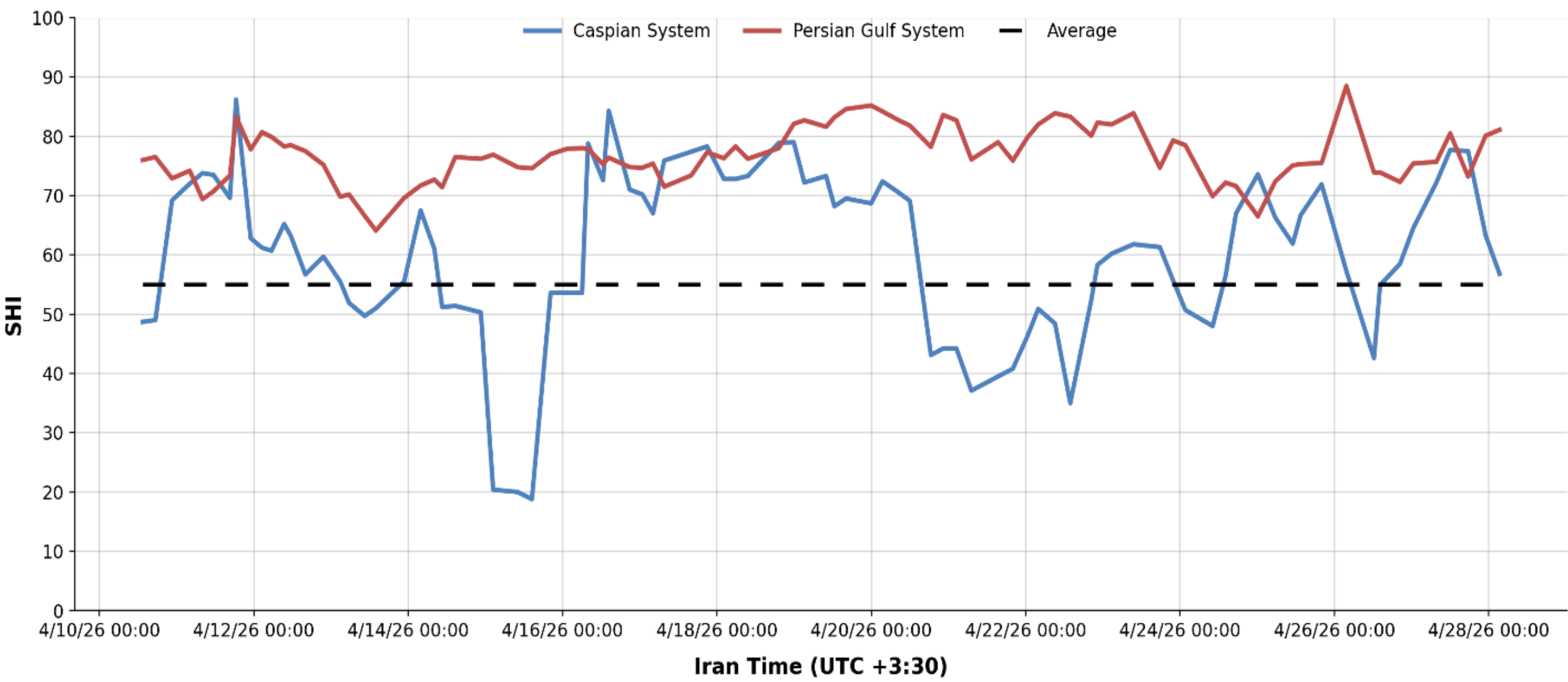


Figure 1. Regional System Health Index by Time, April 10–28, 2026

Caption: The figure compares the time-varying SHI of the Caspian System and the Persian Gulf System from April 10 to April 28, 2026, using Iran time (UTC +3:30), with a reference average of 55.

Source: Author’s monitoring dataset, based on compiled time-series observations of the Caspian System and Persian Gulf System.

Note: SHI is a composite indicator of regional logistics-system health. Higher values indicate stronger system performance and resilience. The dashed line at 55 marks the reference average used for comparison across the observation period.

6.2 Central Asian rail.

Central Asian rail data provide weaker but useful evidence of overland redundancy. China–Central Asia rail and TEU indicators through March 2026 show usable regional capacity, but they do not identify final cargo destination. The evidence therefore supports corridor readiness and potential rerouting capacity, not confirmed Iran-bound deliveries.

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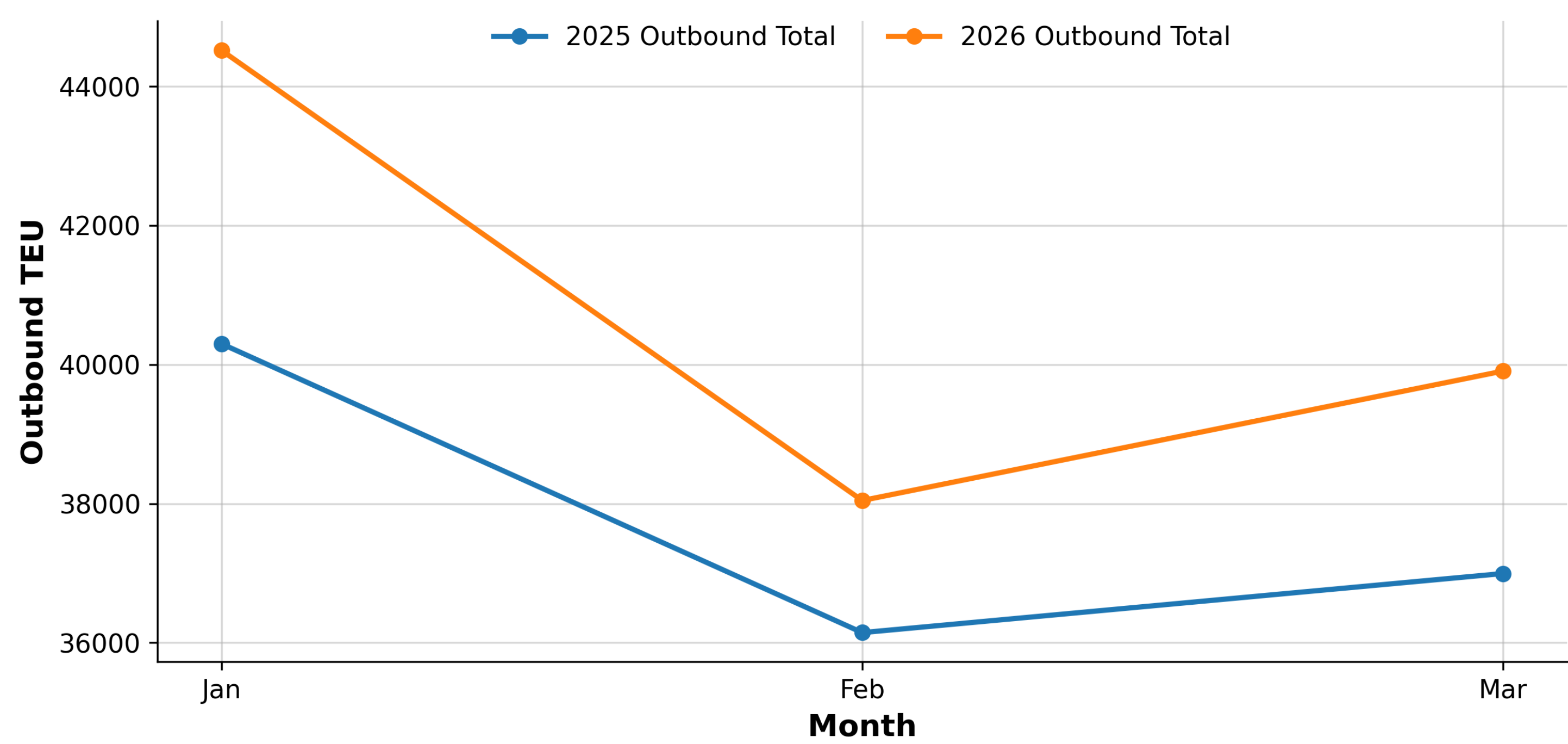


Figure 2. China–Central Asia Rail Activity, January–March 2025/2026 (TEU)

Caption: The figure compares monthly outbound TEU totals for January–March 2025 and 2026, showing that China–Central Asia rail activity remained robust in early 2026.

Source: Author’s compilation based on China Railway Express data, China–Europe Railway Express Portal.

Note: TEU totals indicate corridor scale and overland logistics readiness, not direct evidence of Iran-bound cargo or final destination.

7. The Persian Gulf and Hormuz: Still the Core System

The Persian Gulf and Strait of Hormuz remain Iran’s dominant logistics backbone. Available monitoring data show that Gulf-side throughput still operates at a scale that northern, rail, and overland corridors cannot match.

The ten-corridor framework should therefore be read as a resilience strategy, not a replacement strategy. Alternative routes may slow disruption and preserve minimum flow, but they cannot reproduce Hormuz-centered maritime capacity.

Recent reporting indicates that Hormuz traffic has fallen sharply but not disappeared. This confirms the core pattern: Hormuz remains irreplaceable, but not indispensable for minimum survival. That is the logic of threshold-delaying logistics.

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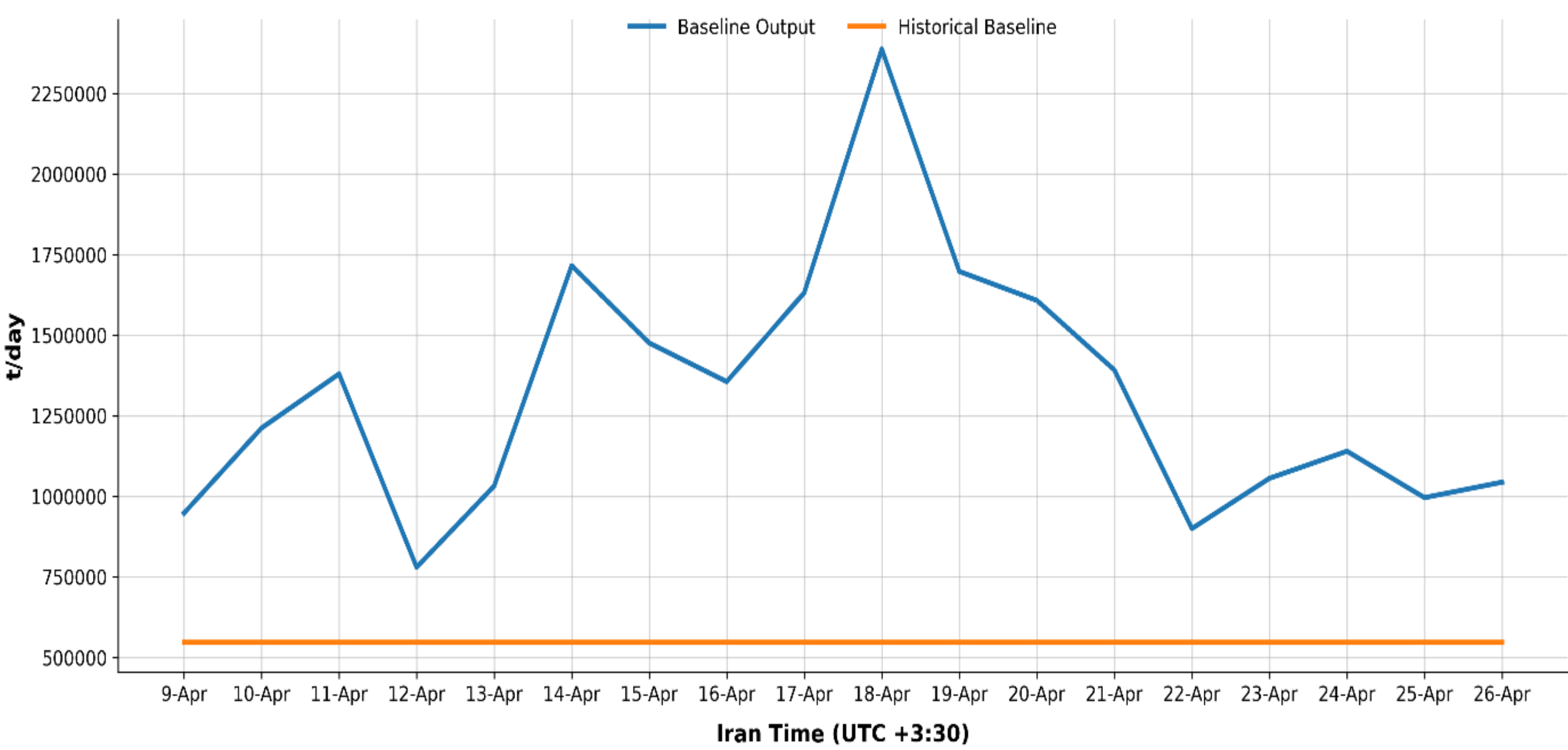


Figure 3. Persian Gulf System Monitor, April 9–26, 2026

Caption: The figure compares estimated daily baseline output in the Persian Gulf system against a historical baseline, showing that Gulf-side throughput remained far larger than northern or overland alternatives despite disruption and volatility.

Source: Author’s Persian Gulf System Monitor dataset.

Note: Baseline output is estimated from observed daily departures using a standardized tons-per-departure assumption. The historical baseline is used as a reference level for comparison, not as a real-time throughput measure.

8. System-Level Interpretation: From Flow Denial to Friction Management

Iran’s ten-corridor adaptation shifts the blockade problem from simple flow denial to friction management. A porous blockade does not fail simply because goods continue to move; it remains effective if throughput falls, delivery times lengthen, costs rise, cargo prioritization weakens, and logistics management absorbs more state capacity and foreign exchange.

Iran’s adaptation therefore changes the form of pressure rather than eliminating it. The contest moves from visible maritime interdiction to distributed logistics friction across ports, rail terminals, customs nodes, border crossings, insurance, finance, vessel availability, air cargo access, and transshipment reliability. The strategic challenge is no longer chokepoint control alone, but network degradation.

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9. Policy Implications

- **Monitoring should shift from ships to systems.** Analysts should track port output, vessel mix, rail activity, border flows, air-route restoration, customs delays, cargo prioritization, and regional price signals—not individual vessels alone.
- **Caspian nodes deserve priority attention.** Northern Iranian ports, Russian Caspian cargo flows, Kazakh and Turkmen transshipment routes, and Caspian vessel availability are key indicators of Iran’s resilience under blockade.
- **Rail and air data should be interpreted carefully.** Central Asian rail activity indicates corridor readiness, not confirmed Iran-bound cargo; air routes indicate selective high-value replenishment capacity, not bulk trade recovery.
- **Pressure should focus on bottlenecks, not whole corridors.** Port berths, rail terminals, customs interfaces, border crossings, fuel access, and aviation scheduling nodes may produce disproportionate system effects.
- **Porous does not mean ineffective.** Residual Hormuz movement and alternative flows do not negate blockade pressure if throughput falls, costs rise, delivery slows, and Iran is forced into inefficient route substitution.

10. Limitations

This brief has three limitations. First, Iran has not released the full ten-corridor list, so the framework here is an analytical reconstruction, not an official map.

Second, the evidence base is uneven. Caspian and Persian Gulf data provide stronger near-real-time indicators, while Central Asian rail data are monthly, available only through March, and do not confirm final destination.

Third, the brief cannot verify cargo composition. Aviation data show connectivity, not specific goods, and the analysis remains a system-level logistics assessment rather than a full cargo audit.

Conclusion

Iran’s ten-corridor plan should not be read as evidence that the country can replace the Persian Gulf logistics system. It should be read as evidence that Tehran is adapting to blockade pressure by building a distributed, lower-capacity, threshold-delaying supply architecture.

The Caspian Sea corridor provides the strongest observable evidence of this adaptation, while Central Asian rail offers a secondary enabling layer. Air connectivity adds a high-value replenishment channel, and residual Hormuz movement shows that the core maritime system is degraded but not fully sealed.

Together, these routes do not restore normal trade. They preserve minimum viable flow. The contest is no longer only over ships passing through a chokepoint, but over whether a dispersed network of ports, rail lines, roads, borders, air routes, and residual maritime windows can keep Iran above the threshold of systemic breakdown.

Iran’s ten-corridor adaptation does not end the blockade problem. It turns it into a long-duration contest of logistics friction.